

DON'T (JUST)
THINK LIKE AN
ARCHITECT!

25 JUNE
—25 JULY 2025

AI HACKATHON
>> 2025



>>TEAM 9

>>TAGFORGE

>>A GENRALIZED
MULTI-AGENT AI
CLASSIFIER FOR CROSS-
PLATFORM BIM IFC
TAGGING

AI HACKATHON
>> 2025

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LEED AP ID+C, LEED AP O+M, WELL AP, RESET AP, BCA GMAP

Principal, Generative AI Agency
Associate Director, Design Integration, Simulation & Technology
M Moser Associates

Jason has spent over 11 years at M Moser advancing innovation in BIM, design tech, and AI. He began as a VDC Designer, delivering complex workplace projects while supporting OSHA and OHSAS 18001 (now ISO 45001) audits.

In 2017, he founded the VRBIM team and launched a SketchUp-based OpenBIM practice, pioneering IFC 2x3 workflows from 2D to 5D across APAC and North America. As Associate, Organisational Development, he scaled BIM operations, standards, and training globally.

In 2019, he founded the DIST group, redefining proposal strategy and cross-functional delivery. He was appointed WELL Faculty and Advisor to buildingSMART China, integrating wellness and sustainability into digital workflows.

In 2024, he co-founded GAIA, M Moser’s Generative AI Agency, onboarding as OpenAI’s first BE-sector client in Singapore. GAIA now assisting in using AI to empower a wide spectrum of services across M Moser.

Jason advises Trimble, Chaos Group, Epic Games, International WELL Building Institute, and buildingSMART on integration, OpenBIM, AI, and digital transformation.

FORWARD BIM

Design and geometry first,
Information to be appended later.

Model first, or information first?

REVERSE BIM

BIM team start to “copy” design
model into Revit from scratch



Creative Modeling Tools

Polygon/NURBS

Freeform Modeling

- Geometry-driven Process
- Freeform, Illustrative
- All layer control are serving the purpose of design

FORWARD BIM

Design and geometry first,
Information to be appended later.

BIM Model Authoring Tools

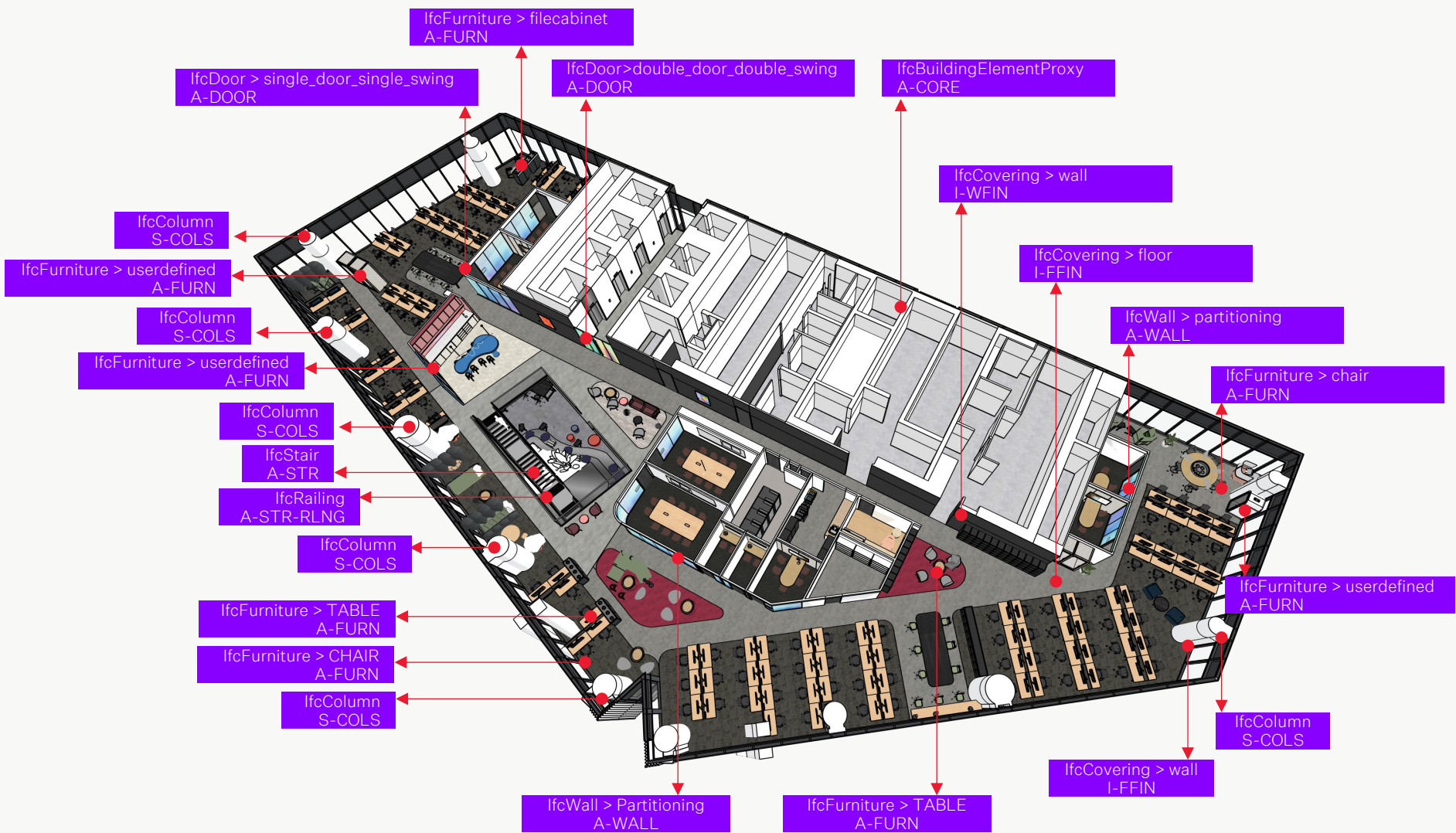
Family/Instance/Attributes

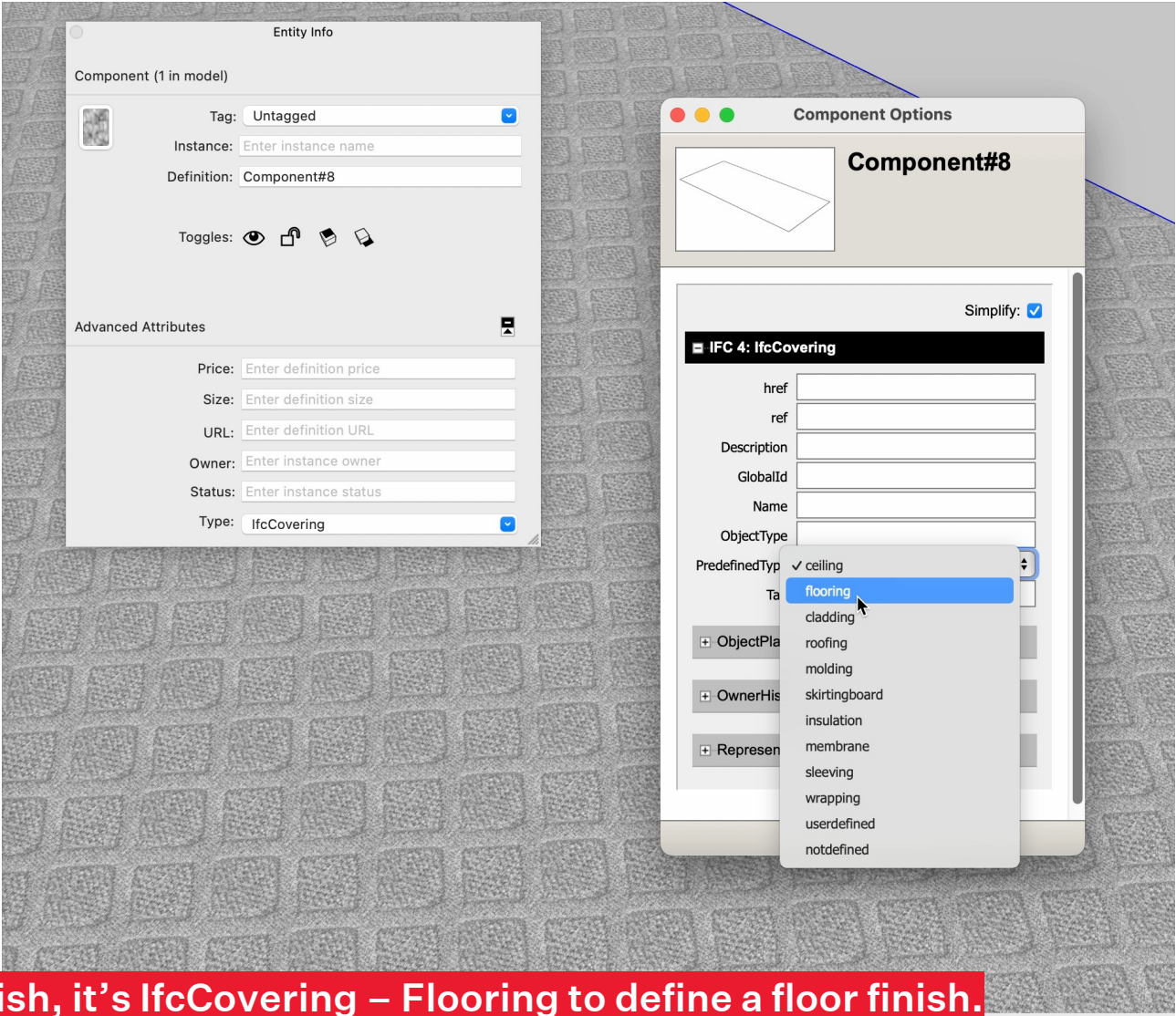
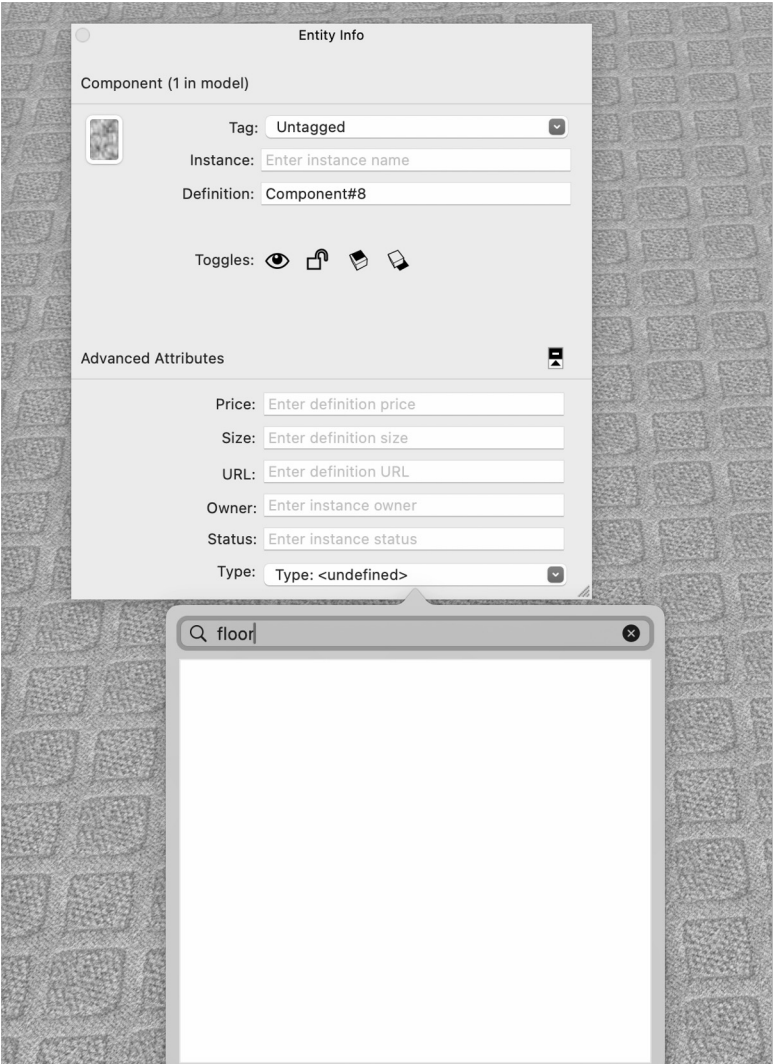
Object-based Modeling

- Information-Drive Process
- Structured, Object-based
- Classification, Types, Attributes are vital

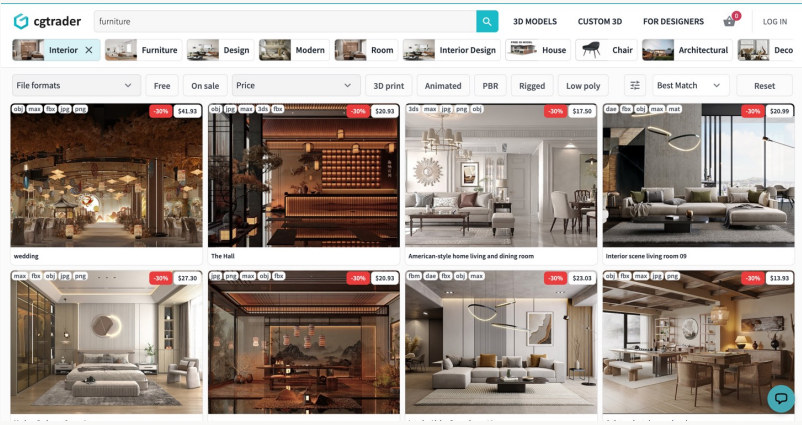
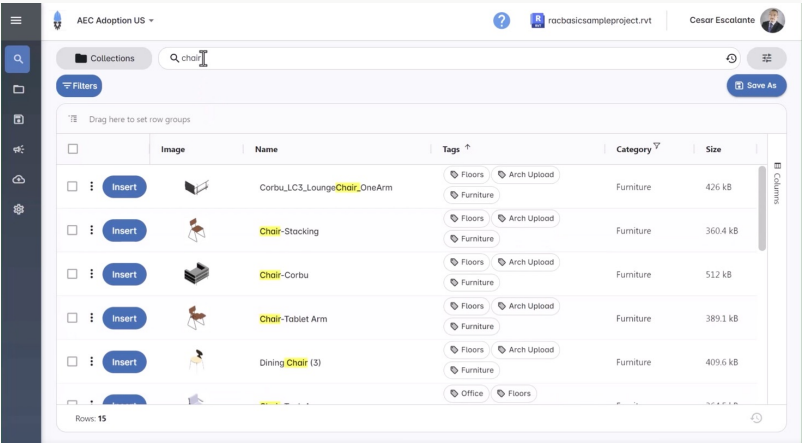
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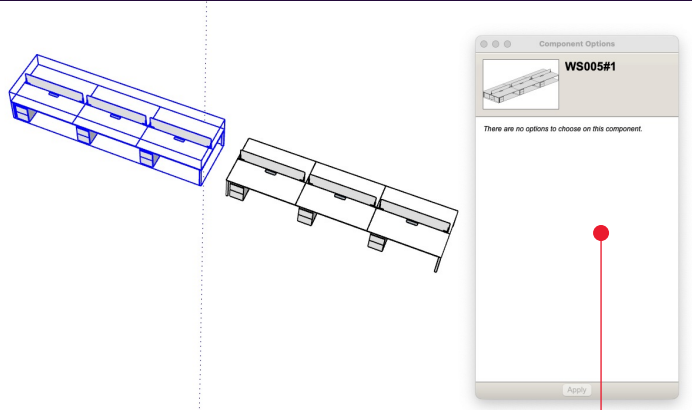


For a Floor Finish, there won't be IfcFloor or IfcFinish, it's IfcCovering – Flooring to define a floor finish.
This is just a simple example of why BIM experience matters, and the accuracy of information – relies on people.

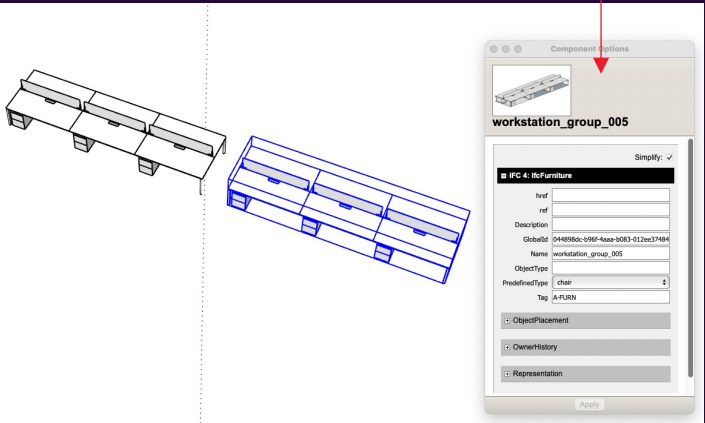


One hand, we create tons of libraries – but nobody really using them

On the other hand, architects and designers are frequently using assets that's not curated from 3rd parties.



ONE CLICK TO BE READY FOR BIM



Objective

ONE-CLICK TO BE READY FOR BIM

Mimic the real-world decision-making of BIM specialists:
To classify and to map components with intelligence.

Scope

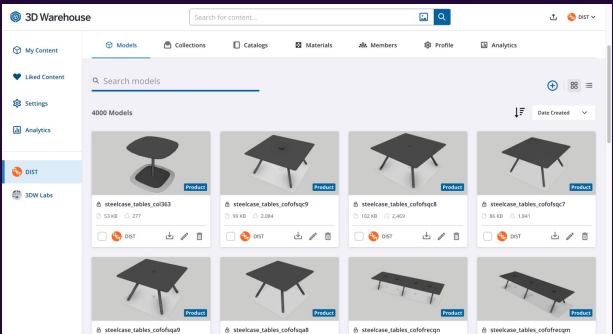
AGENTS TO WORK JUST LIKE A REAL BIM TEAM

Visually interpret the object -> Identify its property -> Match with IFC schema based on practice -> Assign classification and data in SketchUp

Performance Benchmark

ACCURACY, TIME

In library-wide batch processing (the most autonomous scenario), we target $\geq 90\%$ classification accuracy, with average object processing time < 18 seconds.

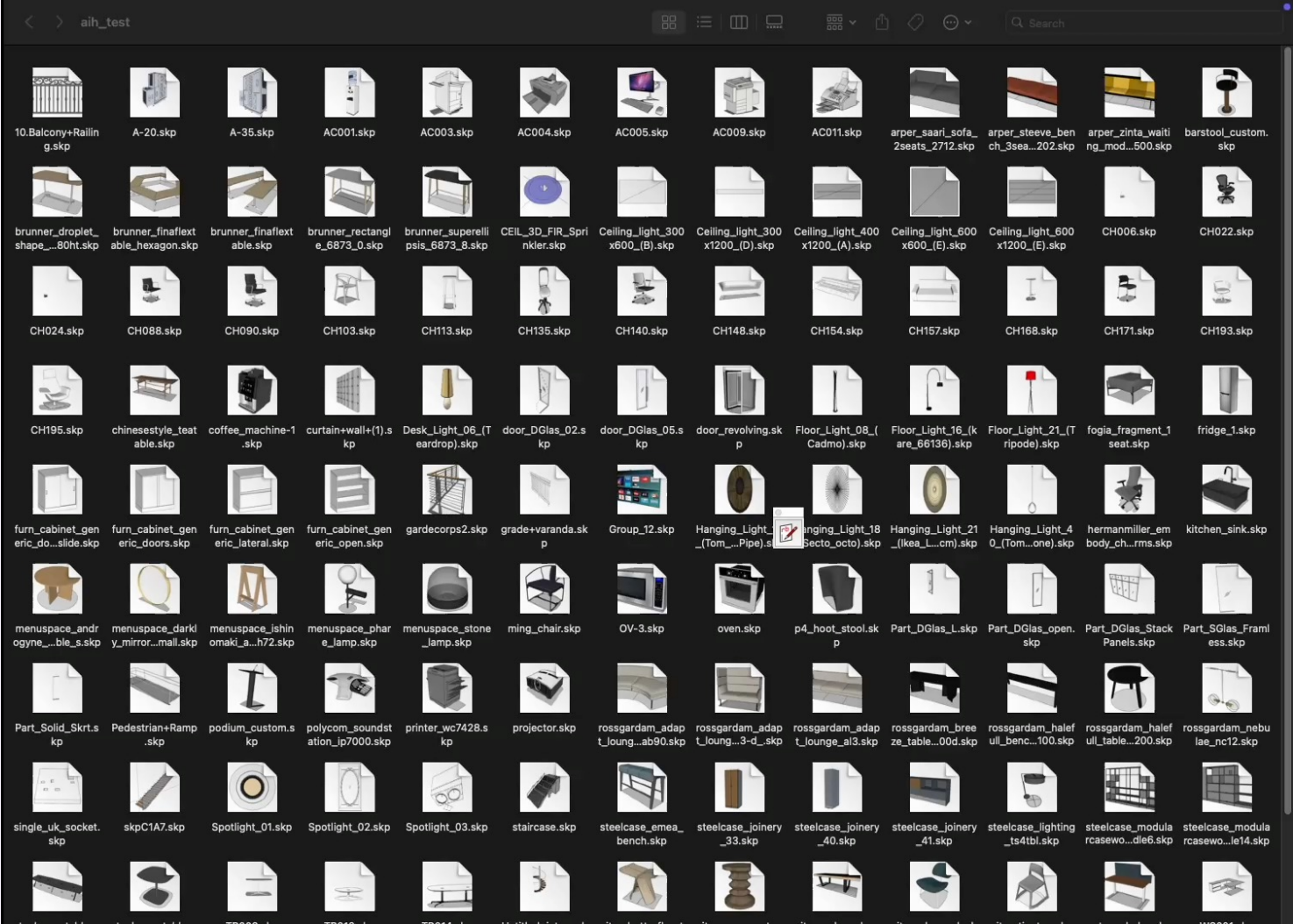


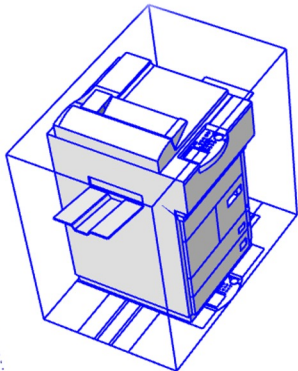
How we test?

- Ran the agentic workflow within SketchUp **over 124 .skp files mixed sources** from M Moser In-house Library, SketchUp 3D Warhouse, SketchFab, etc. comes in various types:
 - **Furniture: Table, Chair, Workstations, etc.**
 - **Lighting Fixtures: Floor Lamp, Spot Lights, etc.**
 - **Equipment: Drinking Water Machine, Fridge, etc.**
 - **Partitions, Doors, Door Frames, etc.**
- It shall automatically assigned:
 - **IFC Classifications (based on IFC4 Schema)**
(e.g. IfcFurniture, IfcElectricAppliance, etc.)
 - **Standardized Layers**
(e.g., A-FURN, E-POWR-APPL)
AIA (American Institute of Architects) Standard.
 - **Attributes** like GUID (Global Unique ID),
PredefineTypes for specific IFC Classes
- Save updated .skp files with IFC-ready metadata

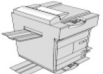
Key Metrics Capturing during the test

- Time per component logging
- Accuracy (comparing with Manual works)
- Error logging (e.g., for missing mappings)





Component Options

 **AC003**

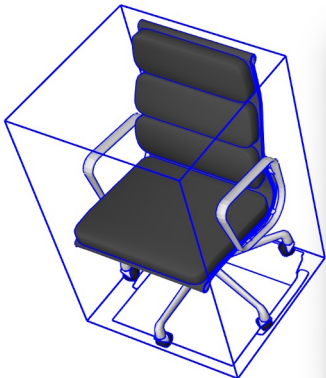
Simplify: ☒

IFC 4: IfcElectricAppliance

href
ref
Description
GlobalId a761b9e4-2faf-47e8-8393-bf6b9dcb2f9
Name AC003
ObjectType
PredefinedType userdefined
Tag E-POWER-APPL

ObjectPlacement
OwnerHistory
Representation

Apply



Component Options

 **CH090**

Simplify: ☒

IFC 4: IfcFurniture

href
ref
Description
GlobalId fcd71e8b-bf37-4aca-94ef-16ef2b7ba4b
Name chair_090
ObjectType
PredefinedType chair
Tag A-FURN

ObjectPlacement
OwnerHistory
Representation

Apply

Component Options

 **Double Glazed Partition framed corner**

adjust other sizes...

partition thickness : pick a standard dimension from list

frame sizes and glass width : type in number followed by unit if not mm, i.e. 2" or 1.3m

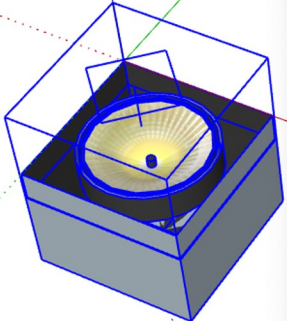
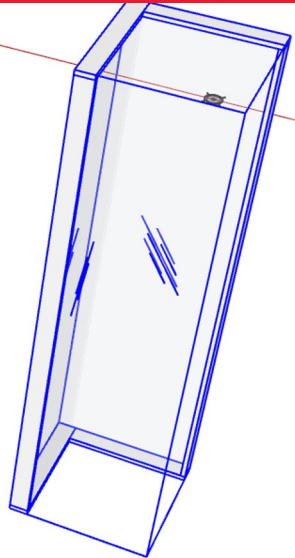
Partition thickness 100 mm
Top / bottom frame width 25 mm
Side frame width 25 mm
Maximum glass panel width 1200 mm

Simplify: ☒

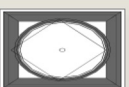
IFC 4: IfcWall

Description
Name Part_DGlas_L
ObjectType
PredefinedType partitioning
Tag A-WALL

Apply



Component Options

 **Spotlight_02**

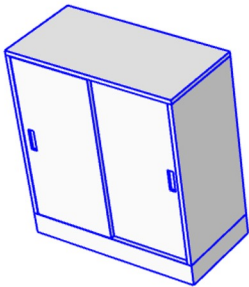
Simplify: ☒

IFC 4: IfcLightFixture

href
ref
Description
GlobalId a2d19d2e-688d-4abf-8081-1b8b293425
Name Spotlight_02
ObjectType
PredefinedType pointsource
Tag E-LITE-FIXT

ObjectPlacement
OwnerHistory
Representation

Apply



Component Options

 **Cabinet with open shelves**

Simplify: ☒

IFC 4: IfcFurniture

href
ref
Description
GlobalId 1047ddc2-7e03-4d04-ac68-6c6668015
Name furn_cabinet_generic_doors_slide
ObjectType
PredefinedType filecabinet
Tag A-FURN

ObjectPlacement
OwnerHistory
Representation

Apply

Component Options

 **Solid door with level handles**

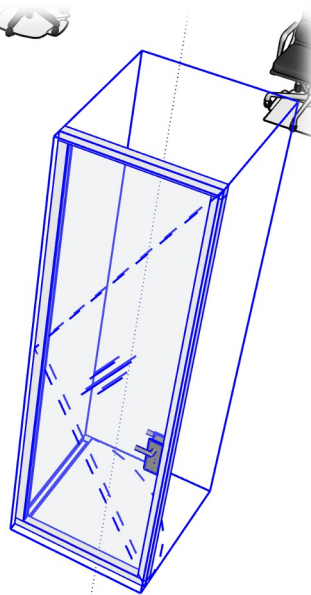
Door leaf thickness 50 mm
Top / bottom frame width 50 mm
Side frame width 50 mm
Height to handle center 1000 mm
Maximum pane width 1200 mm

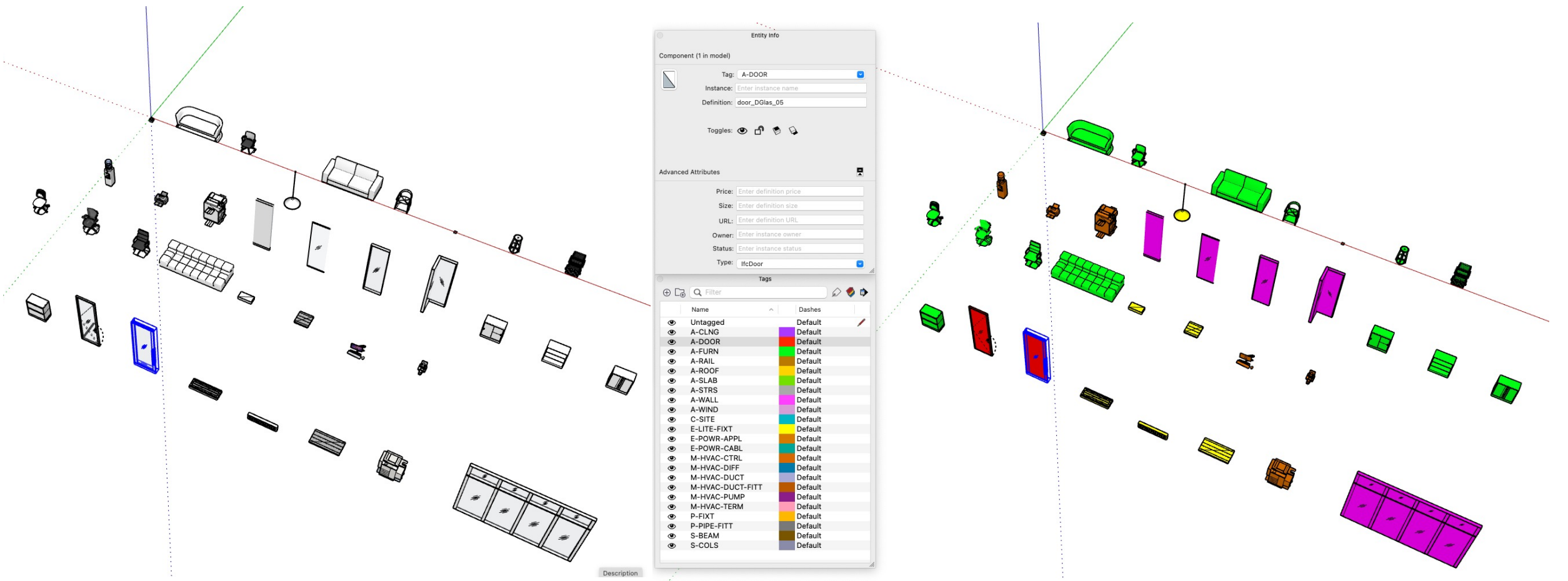
Simplify: ☒

IFC 4: IfcDoor

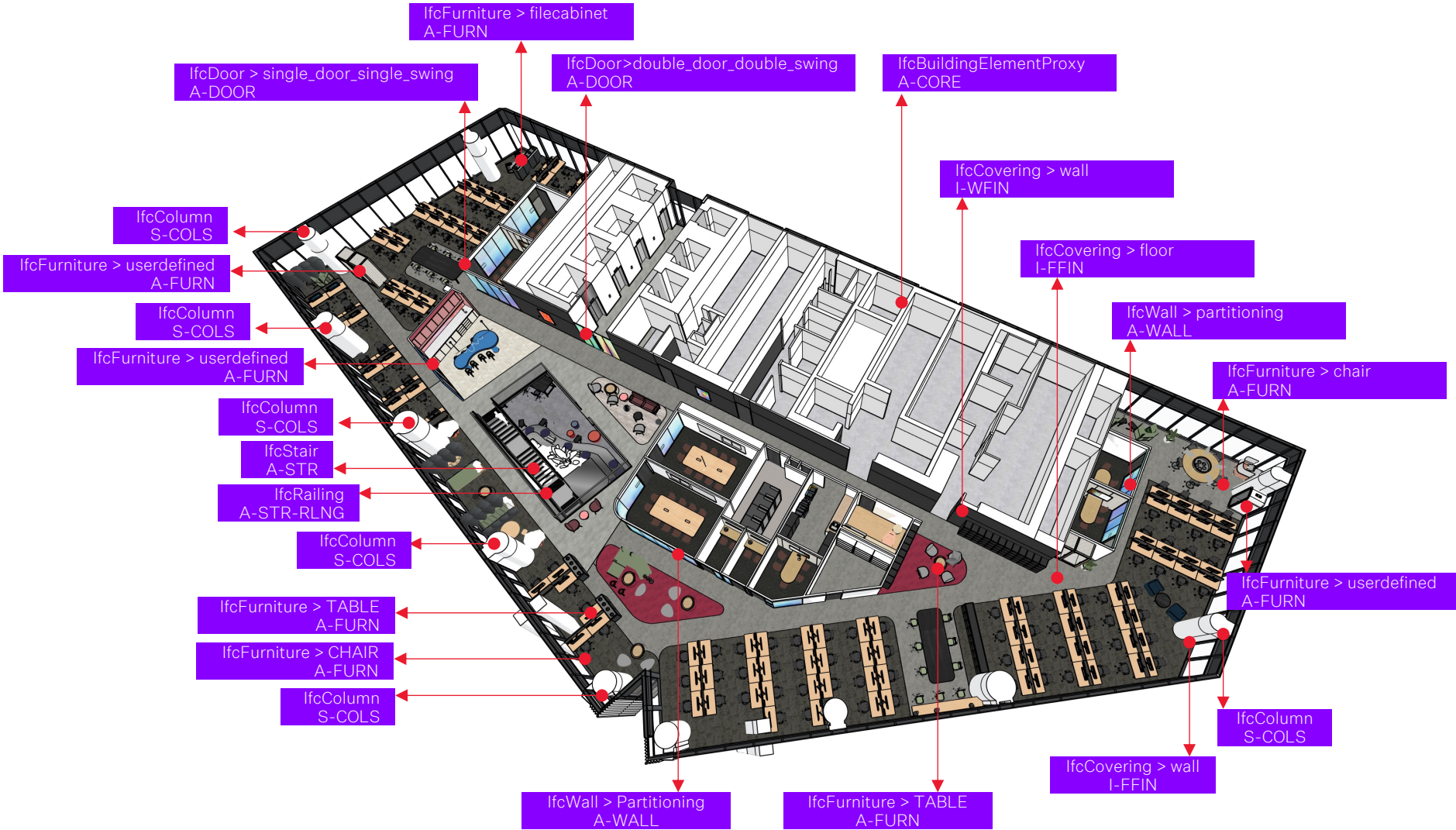
Description
Name door_DGlas_02
ObjectType
OperationType single_swing_left
OverallHeight 2.7
OverallWidth 0.9
PredefinedType door
Tag A-DOOR
UserDefinedOperationType

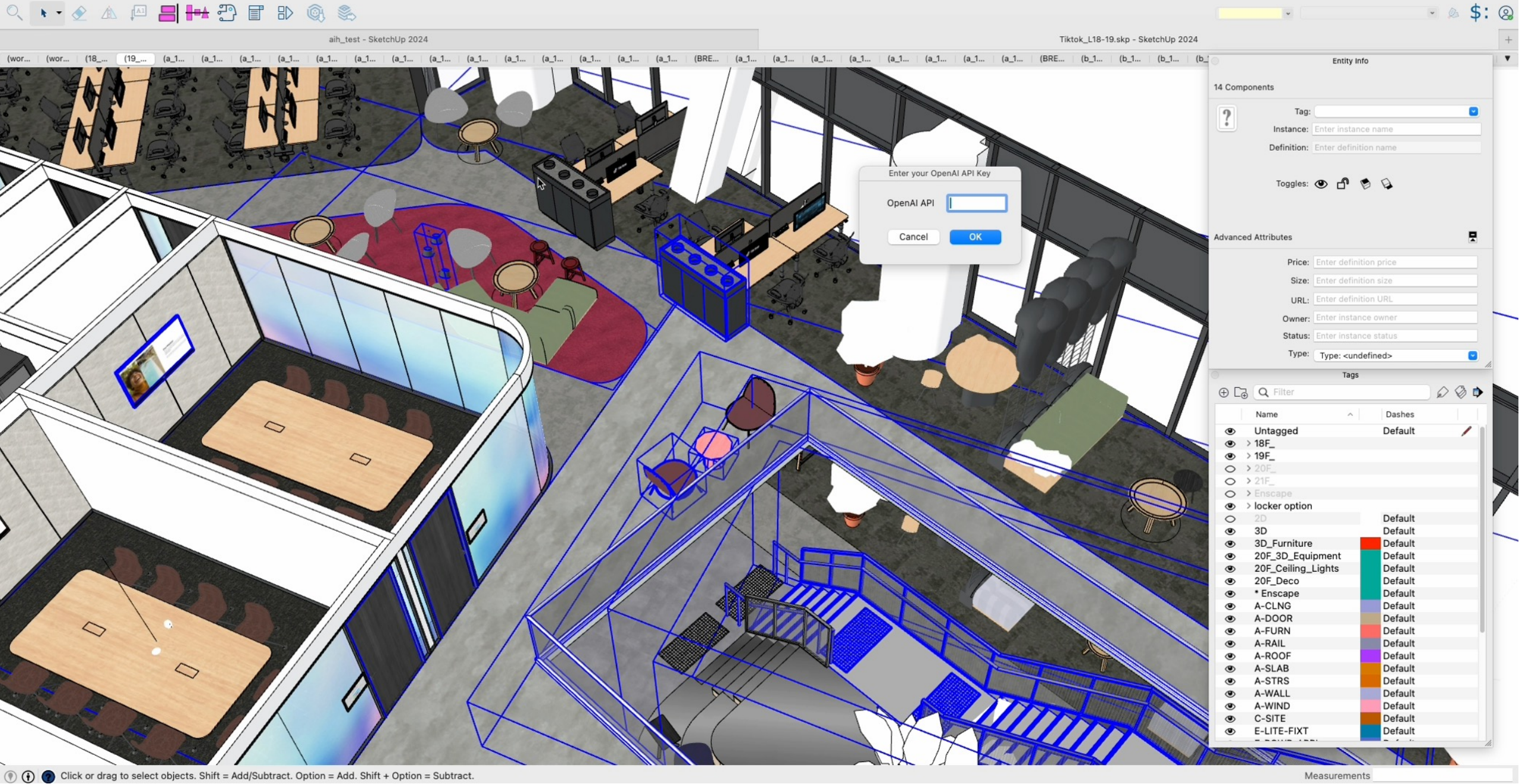
Apply

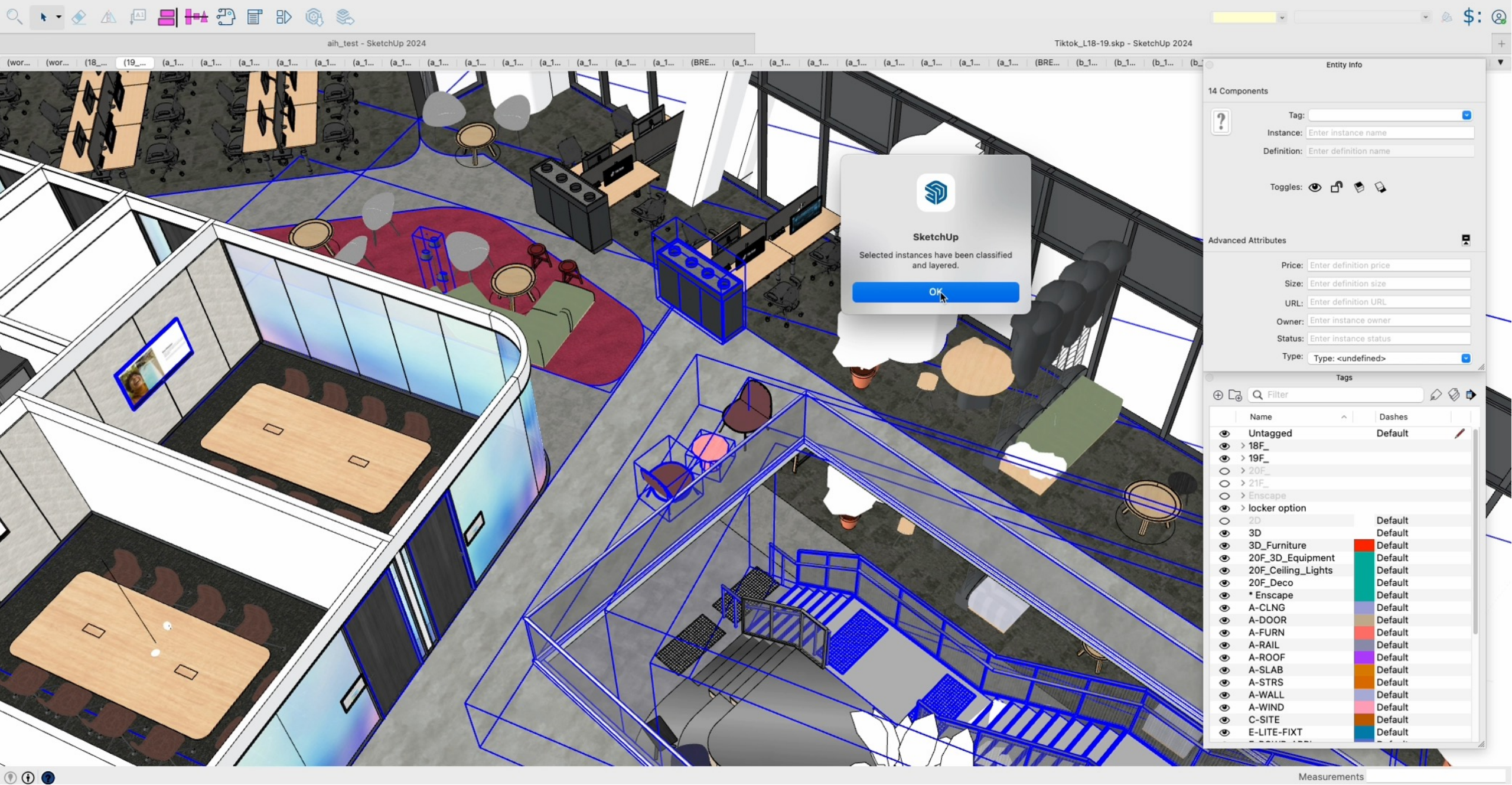


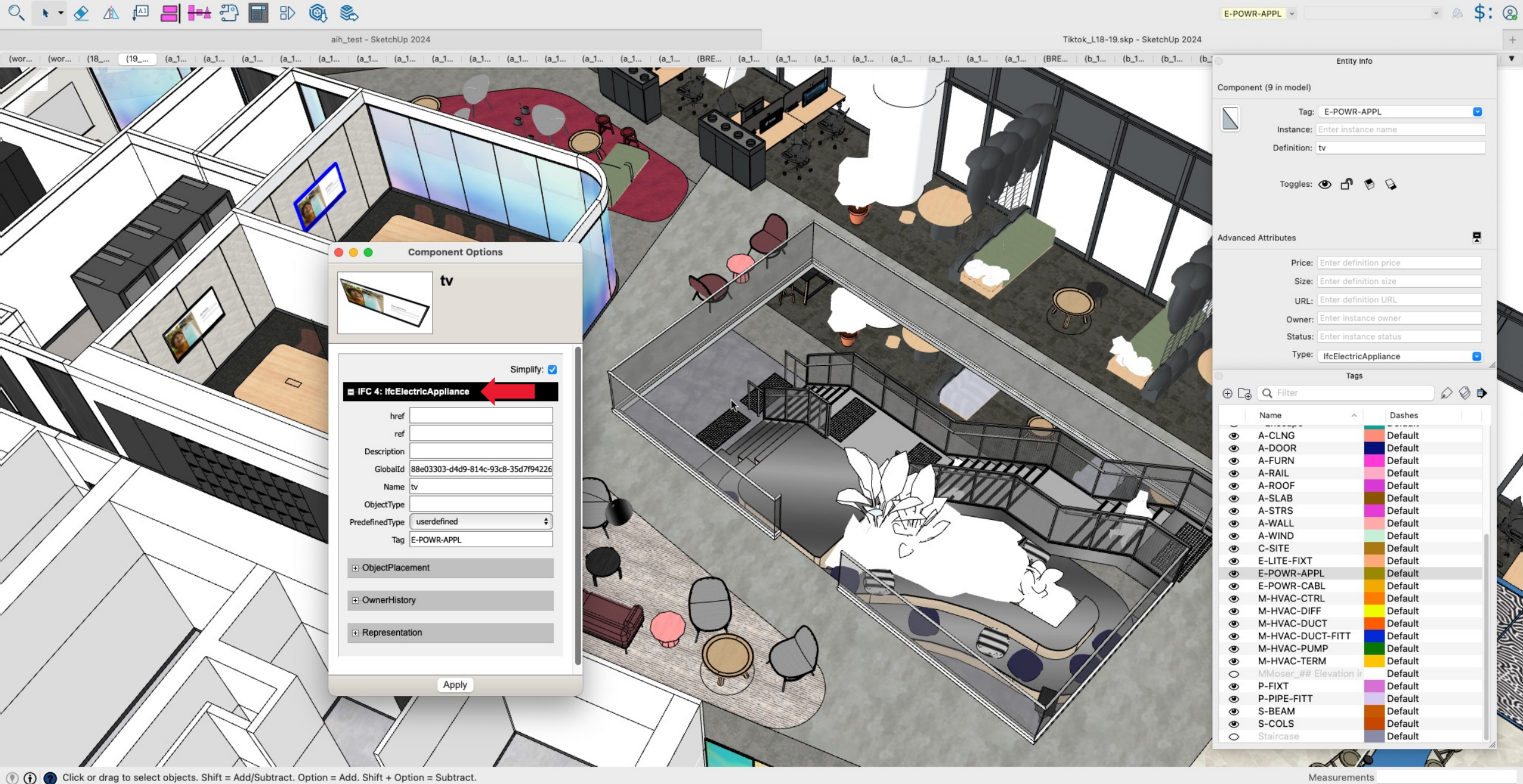


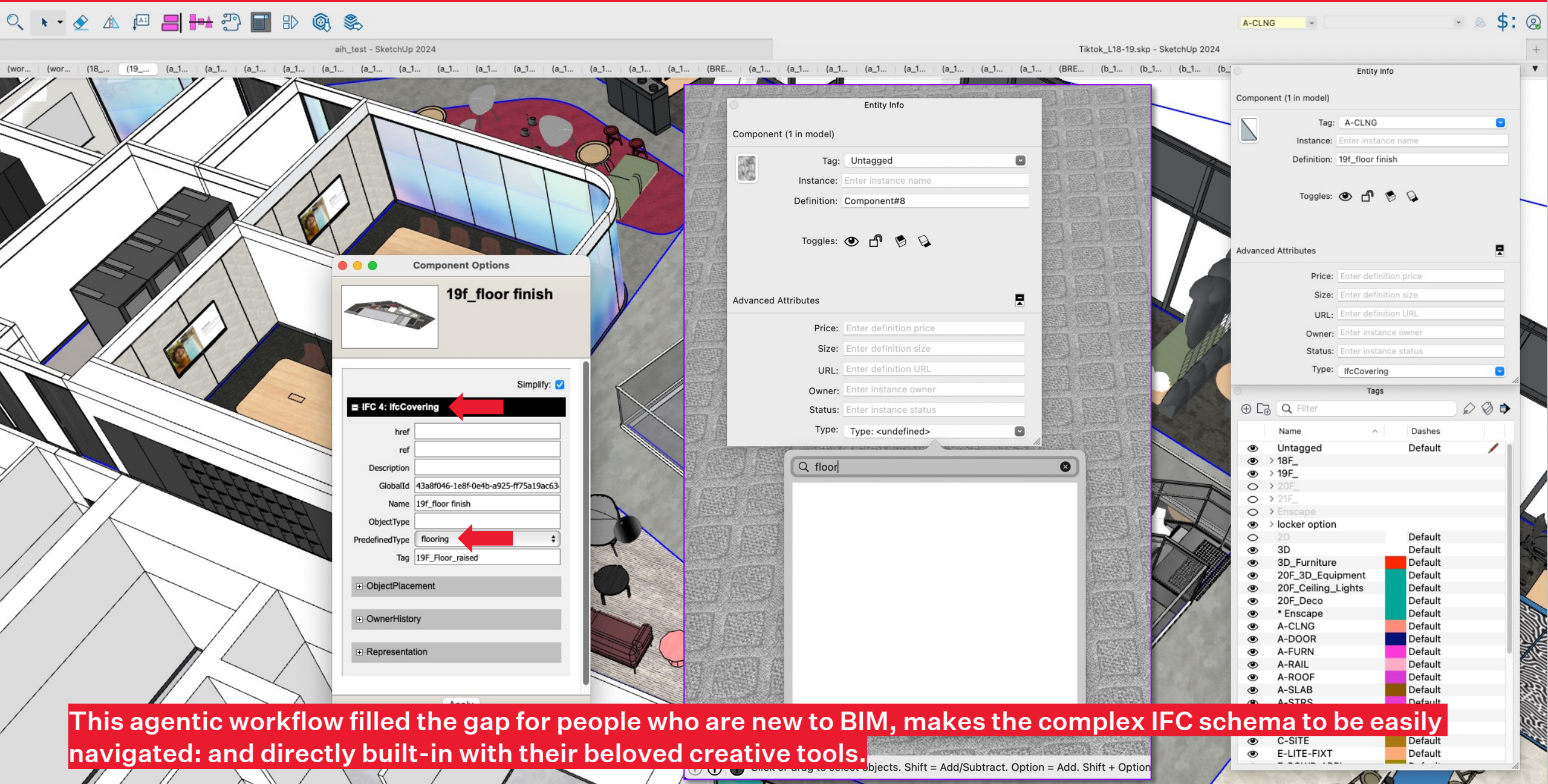
Layers/Tags of individual objects also can be prompted by our agentic workflow as well.



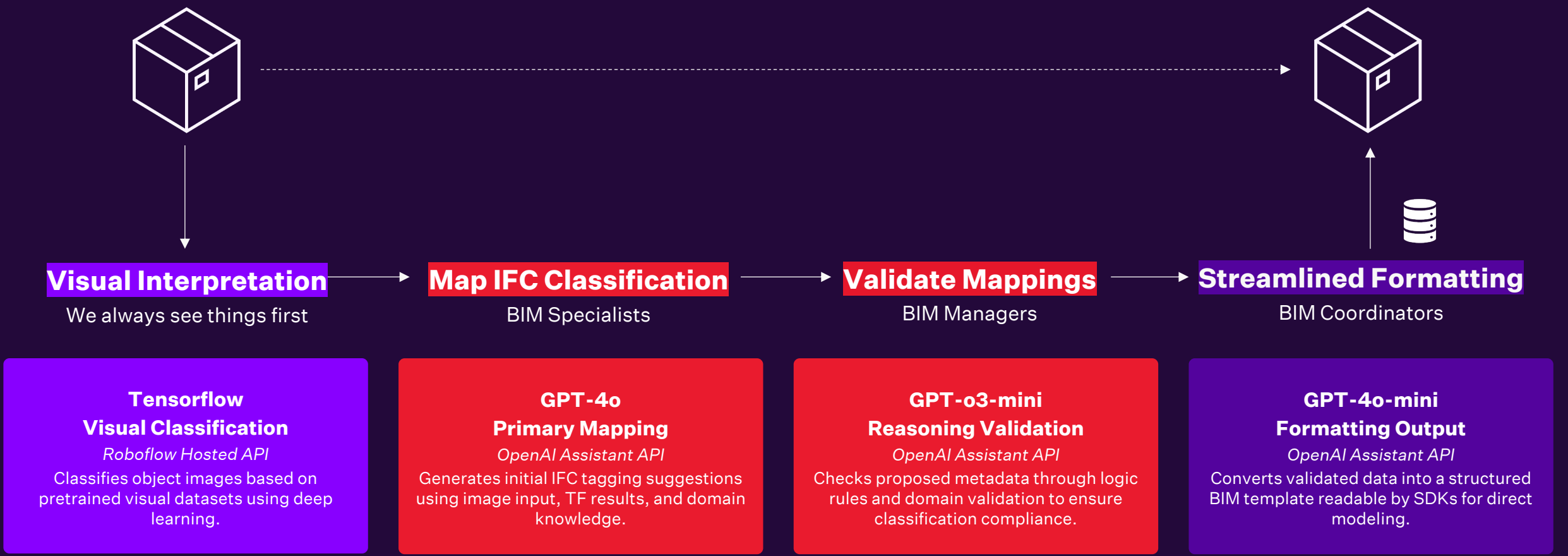






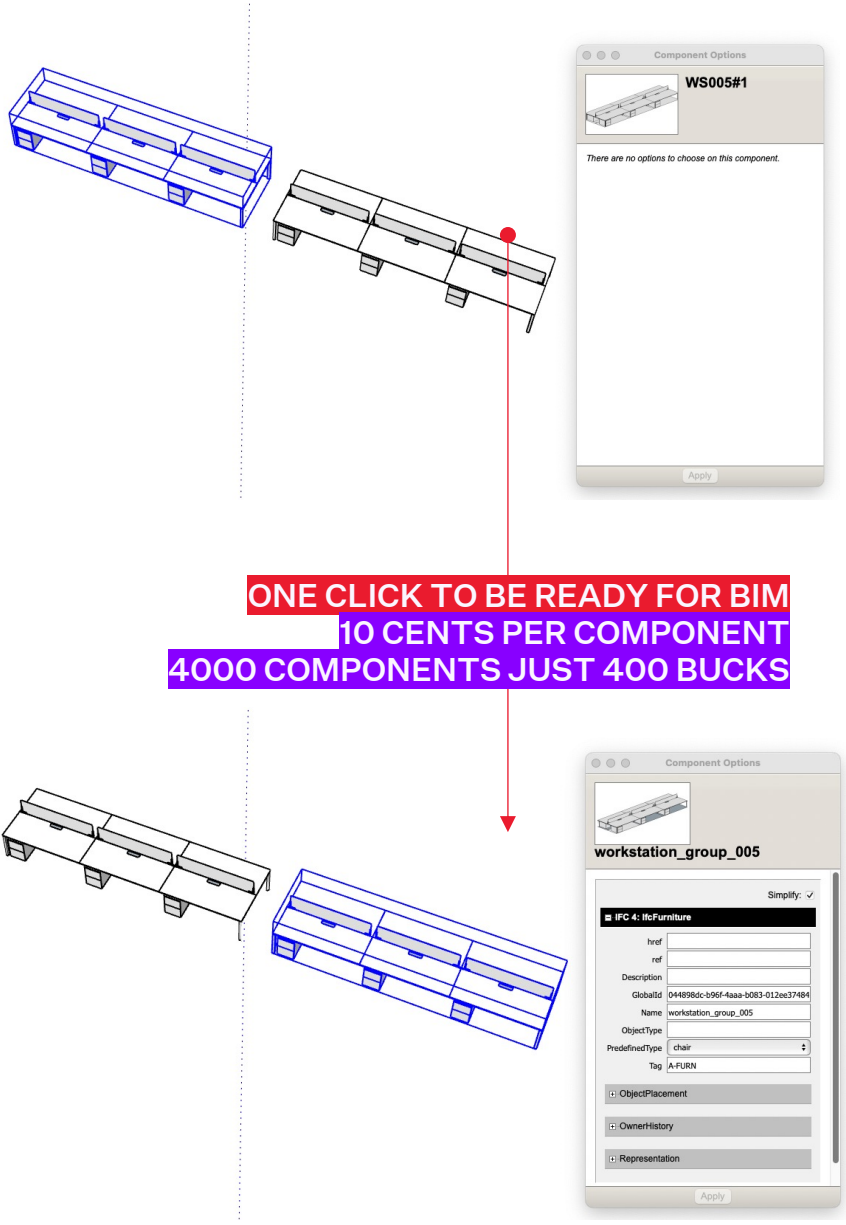


This agentic workflow filled the gap for people who are new to BIM, makes the complex IFC schema to be easily navigated: and directly built-in with their beloved creative tools.



Result Metric	Expectation	Performance	Notes
Classification Accuracy	>=90%	95.16%	+5.16%, 118/124 Files
Layer Accuracy	>=90%	93.55%	+93.55%, 116/124 Files
Time per Component	<20 sec	17.91 secs	Total 2221 secs
Max Observed Time	N/A	30.13 secs	On CH024.skp only

Service	Usage	Unit Rate (\$)	Cost / Component
Roboflow Inference	18 API calls per component	\$0.003 per inference	\$0.054
GPT-4o	1,500 tokens (input + output)	\$0.005/input 1K tokens, \$0.015/output 1K tokens	\$0.015
GPT-o3-mini-high	2,868 tokens (input + output)	\$0.15/input 1M tokens, \$0.60/output 1M tokens	\$0.0011
GPT-4o-mini	7,726 tokens (input + output)	\$0.15/input 1M tokens, \$0.60/output 1M tokens	\$0.0031
Sub-total Cost per Component (USD)			\$0.073
Sub-total Cost per Component (SGD)			\$0.095



The diagram illustrates a workflow for BIM component configuration. It starts with a 3D model of a workstation group, which is then processed through a 'Component Options' window. A red arrow points from the 3D model to the 'Component Options' window, which displays the component name 'WS005#1' and a message 'There are no options to choose on this component.' Below this, a red arrow points to another 'Component Options' window, which displays the component name 'workstation_group_005' and a list of properties including 'href', 'ref', 'Description', 'GlobalId', 'Name', 'ObjectType', 'PredefinedType', and 'Tag'. The 'ObjectType' is set to 'chair' and the 'Tag' is 'A-FURN'. The 'Component Options' window also includes a 'Simplify' checkbox and an 'Apply' button.

ONE CLICK TO BE READY FOR BIM
10 CENTS PER COMPONENT
4000 COMPONENTS JUST 400 BUCKS

Visual Interpretation

We always see things first

**Tensorflow
Visual Classification**

Roboflow Hosted API

Classifies object images based on pretrained visual datasets using deep learning.

Map IFC Classification

BIM Specialists

**GPT-4o
Primary Mapping**

OpenAI Assistant API

Generates initial IFC tagging suggestions using image input, TF results, and domain knowledge.

Validate Mappings

BIM Managers

**GPT-o3-mini
Reasoning Validation**

OpenAI Assistant API

Checks proposed metadata through logic rules and domain validation to ensure classification compliance.

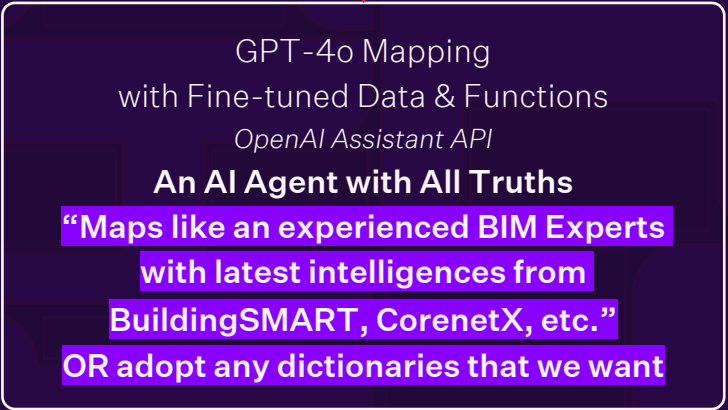
Streamlined Formatting

BIM Coordinators

**GPT-4o-mini
Formatting Output**

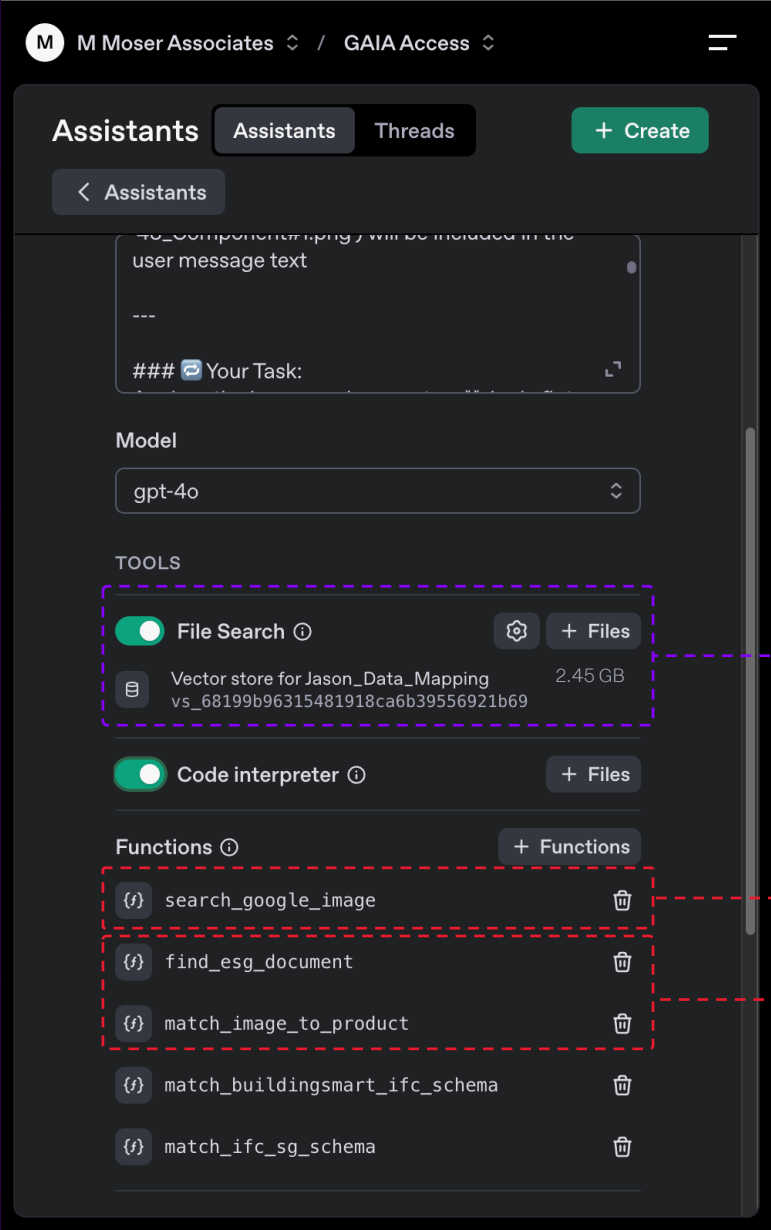
OpenAI Assistant API

Converts validated data into a structured BIM template readable by SDKs for direct modeling.



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Computerize Image to JSON

Conversion of Visual Tagged Datasets as json and serve as GPT's fine-tune data

Function: Image Search & Matching

Use mature engines (e.g. Google Image Search with Exact filters)

Function: Multi-shot Web Search

Attempt multiple times using different mature search engine to validate the recognition result

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Result Metric	Expectation	1.0 Performance	2.0 Performance	Performance Improvements
Classification Accuracy	>=90%	(118/124) 95.16%	99.54% (219/220)	+4.6% Classification Accuracy
Layer Accuracy	>=90%	(116/124) 93.55%	99.54% (219/220)	+6.4% Layer Accuracy
Time per Component	<20 sec	17.91	8.22	2.17x Faster
Max Observed Time	N/A	30.13	11.21	2.68x Faster

Service	Usage	Unit Rate (\$)	Cost / Component
Roboflow Inference	18 API calls per component	\$0.003 per inference	\$0.054
GPT-4o	1,500 tokens (input + output)	\$0.005/input 1K tokens, \$0.015/output 1K tokens	\$0.015
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Sub-total Cost per Component (SGD)			\$0.095

M Moser Associates / GAIA Access

Usage View legacy dashboard Export

GAIA Access 05/13/25 - 05/13/25

Total Spend \$1.16

Total tokens 667,562

Total requests 220

0

AC +/- % ÷

7 8 9 ×

4 5 6

1 2 3

0 . =

219/220 Accuracy

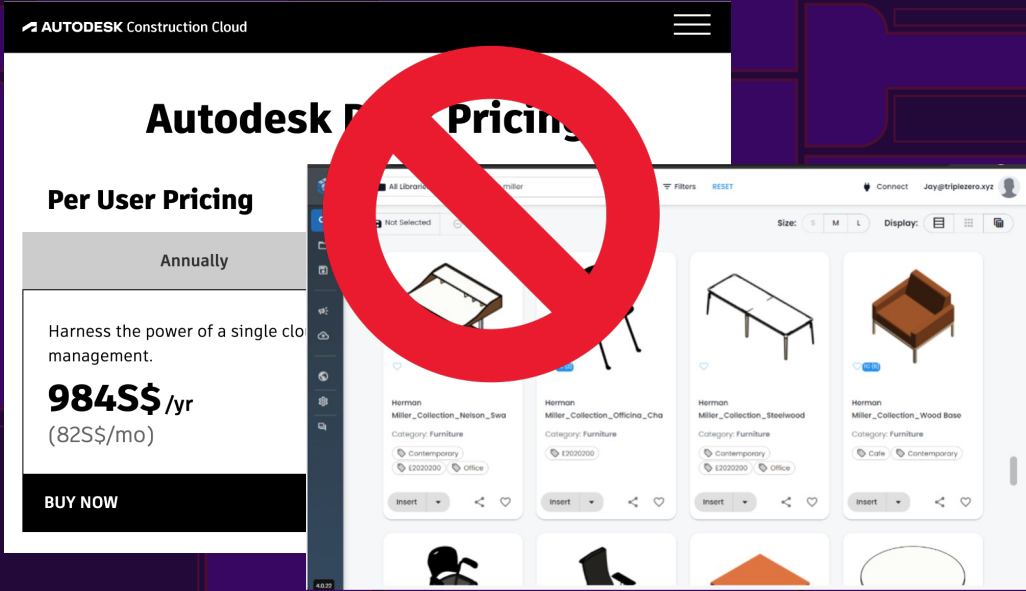
2.2x faster

13.7x cheaper

*Based on OpenAI Enterprise Plan

WHAT'S NEXT?

- Plugins with More Tools**
We're expanding beyond SketchUp — soon this system can run in Revit, Rhino, and ArchiCAD or WebGL APIs.
- Expanding to Architecture, Engineering etc.**
Columns, Slabs, Curtain Walls, or FCU, VAV, Sprinkler Heads and Power Sockets? Your call.
- From POC to Full Platform + Plugins**
This started as a proof-of-concept. And we've built the SketchUp Plugin for I/O with our agentic workflow: Next, we'll go through programs hosted by industry firms to turn it into a full system that helps you all to manage your 3D libraries automatically.



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I wish this tool will help us and remind us our original mission:
to create, to connect, and to care.

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SINGAPORE
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